

CURRICULUM VITAE

JANE DOE

Born January 1, 1990
Example citizen

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RESEARCHER in Statistics
Example Institute
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BRIEF SUMMARY OF ACTIVITIES

Research	STATISTICAL METHODS, MACHINE LEARNING
Themes	Sparse models · Multi-omics integration · Reproducible research
Production	2 papers, 2 talks
Students	1 PhD, 1 Master
Teaching	Statistics in Action with R, Convex Optimization

EXPERIENCE

2011–	RESEARCHER, EXAMPLE INSTITUTE
2008–2011	POSTDOC, EXAMPLE UNIVERSITY
supervisor	Dr. Example

EDUCATION

2015	HABILITATION IN MATHEMATICS
Title	<i>Sparse methods for complex data</i>
2003	PHD IN APPLIED MATHEMATICS

SCIENTIFIC ACTIVITIES

GRANTS

ON GOING PROJECTS

2023–2027	DISCERN – DISCOVERING THE CAUSES OF A POORLY UNDERSTOOD CANCER
Partners	20 partners
Support	Horizon Europe
Description	A five-year consortium project combining statistical genomics, imaging, and epidemiological cohort data to identify early biomarkers of a poorly understood cancer subtype. My group leads the statistical modelling work package, developing sparse graphical models for multi-omics integration across the twenty partner sites.

PAST

2019–2023	ECONET – ADVANCED STATISTICAL MODELLING OF ECOLOGICAL NETWORKS
Support	French National Research Agency (ANR)
2017–2019	SEARS – SAMPLING STRATEGIES AND NETWORK ANALYSIS
Leader	Mathieu Thomas

OTHER GRANTS

2010–2012 <i>Support</i>	OLDER GRANT, LISTED ONLY IN THE LONG VERSION French National Research Agency (ANR)
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ACTIVITIES

SCIENTIFIC COMMITTEES

<i>since 2022</i>	COUNCILS Member of the Foundation Council
2020–2024 <i>role</i>	PHD COMMITTEE (full list) Reviewer for 12 PhD theses

STUDENTS

PHD STUDENTS

2021–2024 <i>Co-supervisor</i> <i>Description</i>	JANE ROE – SPARSE STATISTICAL METHODS FOR LARGE-SCALE MULTI-OMICS DATA Prof. Example Developed sparse graphical model estimators for integrating heterogeneous omics data sources, with applications to the DISCERN consortium. Defended in 2024; now a postdoctoral researcher at Example University.
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MASTER STUDENTS

2023 <i>Co-supervisor</i>	JOHN ROE – POISSON-LOGNORMAL MODELS FOR COUNT DATA Dr. Example
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TEACHING

Approximately 2000 hours of teaching. Lecturer at Example University from 2015 to 2024.

2020–24 <i>Msc</i>	STATISTICS IN ACTION WITH R Probabilistic models, data analysis, R programming
2017 <i>Msc</i>	CONVEX OPTIMIZATION (12h course) Gradient methods, Newton method, Proximal methods

SCIENTIFIC PRODUCTIONS

PUBLICATIONS

SELECTED PUBLICATIONS

- [JP1] J. DOE et J. ROE. “A Toy Example for the quarto-cv-blocks Extension”. In : *Journal of Reproducible Examples* 1 (2023), p. 1-10. DOI : [10.0000/example.2023.001](#).
- [JP2] J. DOE. “Demonstrating a Fictitious Conference Paper”. In : *Proceedings of the Example Conference*. 2021, p. 42-50.

SELECTED TALKS

- [CT1] J. DOE. “Reproducible CVs with Quarto”. In : *Example Workshop on Reproducible Documents*. Example City, 2022.
- [CT2] J. DOE et J. ROE. “A Fictitious Talk for the quarto-cv-blocks Example”. In : *Proceedings of the Other Example Conference*. 2020, p. 12-20.

RESEARCH PROJECTS

DISCERN

As part of the DISCERN project, my group develops sparse graphical model estimators for integrating multi-omics data. Given a sample covariance matrix S , we estimate a sparse precision matrix $\hat{\Theta}_\lambda$ by solving

$$\hat{\Theta}_\lambda = \arg \min_{\Theta \succ 0} -\log \det \Theta + \text{tr}(S\Theta) + \lambda \|\Theta\|_1.$$

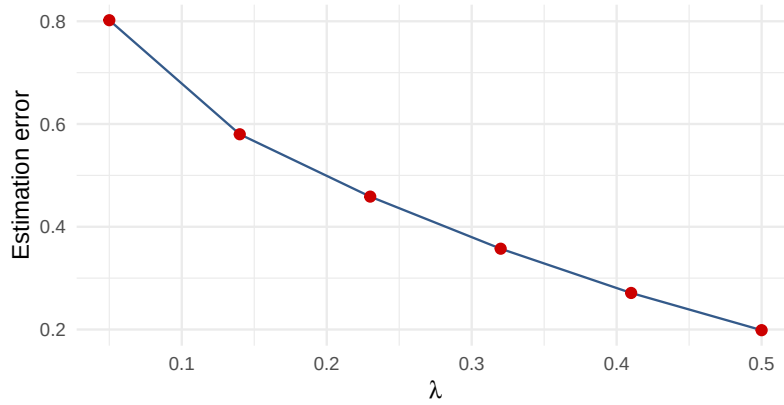


FIGURE 1 : Estimation error as a function of the sparsity penalty

Increasing the penalty λ trades estimation error for a sparser, more interpretable graph, see Figure 1.

ANOTHER PROJECT

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See Table 1.

Lambda	Error	Sparsity
0.05	0.79	12%
0.20	0.42	48%
0.50	0.19	81%

TABLE 1 : Estimation error and graph sparsity for a few penalty values